Cover letter — final author-review draft

Dear Professor Haueisen,

Please consider our Research Article, “Disentangling spatial-filter and classifier contributions to limited-label motor-imagery EEG recalibration”, for Biomedical Engineering / Biomedizinische Technik.

The manuscript addresses how a short calibration run should change a motor-imagery decoder. By giving classifier-only recalibration, full spatial refitting and support-only alignment the same source and current-session labels, we quantify the contribution of each update operation on identical query trials. In the original twelve-participant Kumar cohort, spatial refitting added 5.47 balanced-accuracy points beyond classifier recalibration and accounted for 73.7% of the total mean improvement over freezing. Prediction transitions and five-session participant summaries explain that increment.

The completed calibration-budget analysis adds a second spatial operation. Support-only alignment increased balanced accuracy by 5.30 points beyond full refitting in Kumar (pointwise 95% interval 2.81–8.26) and by 5.65 points in the separately fixed nine-participant BNCI comparison (3.15–8.70; secondary Holm $p=0.01172$). All nine external participants improved. Full versus classifier-only refitting in BNCI differed by 0.19 points (-1.85 to 2.13). Complete budget curves, marker-aware window sensitivity and the full neural/geometric controls accompany these comparisons.

The contribution is a reproducible analysis of where limited-label recalibration gains arise, supported by shared trial identities and participant-level estimates. The study uses our recorded analyses of public datasets; source methods and acquisition studies receive their appropriate citations. Data/code availability, national-law secondary-use ethics and actual AI assistance are documented in the separate declarations.

The author list is Ziyuan Zhu, Sihao Liu and Haoyang Yu, all sharing the Chongqing Medical University affiliation. The authors report no conflicts and no study funding. The prior PLOS ONE submission was rejected, as confirmed by the corresponding author. This revised version is supplied for final author review; ScholarOne submission has not been performed. The existing five suggested reviewers are supplied for the authors to verify against the expanded author list.

Sincerely,

Ziyuan Zhu, corresponding author

College of Artificial Intelligence Medicine, Chongqing Medical University, Jinyun Campus, No. 61 Daxuecheng Middle Road, Shapingba District, Chongqing 401331, China

zzy2630816871@gmail.com

ORCID: https://orcid.org/0009-0005-1153-4926